

# Industrial Phytochemistry & HPTLC Standardization

Learn how a botanical sample moves from receipt to a defensible chromatographic fingerprint.

## Lecture notes

1. Establish identity before testing: botanical name, plant part, supplier, batch and collection record must agree.
2. Make the sample traceable: assign a lab number, record weight, solvent, extraction ratio and analyst.
3. Control the separation: plate type, mobile phase, chamber saturation, application volume and detection wavelength affect reproducibility.
4. Read results cautiously: compare retention behaviour and marker response with a reference standard; document deviations before release.

## Watch with purpose

While watching the YouTube lecture, pause at each workflow decision. In your notes, identify: (a) the evidence needed before acting, (b) the safety, ethics or quality control, and (c) the record that proves the step occurred.

## Knowledge check

| Question                                                         | Expected answer                                                                                      |
|------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| Why is a reference standard run with the sample?                 | It provides a known comparison so the marker identity and response can be evaluated.                 |
| What should happen if a mobile-phase ratio differs from the SOP? | Record the deviation, assess its impact and obtain the required review before relying on the result. |

## Practice MCQs

1. Which choice best supports reliable professional practice?  
A. Relying on memory after the session B. Recording approved checks and escalating deviations C. Skipping documentation when busy D. Changing findings after review
2. What should guide an action in a supervised learning workflow?  
A. A generic online claim B. The approved process, evidence and appropriate oversight C. A marketing target D. An assumption about the case

**Answer key:** 1-B and 2-B. The course emphasises evidence, transparent records and appropriate supervision.

## Applied learning record

Use this page after watching the lecture and reading the notes. It can support a discussion with an industry mentor, faculty supervisor or placement-cell reviewer.

| Reflection prompt                                     | Your notes |
|-------------------------------------------------------|------------|
| Key concept explained in my own words                 |            |
| Evidence or record I would create                     |            |
| Safety, ethics or quality control I would check       |            |
| One question for my mentor                            |            |
| How this relates to an internship or entry-level role |            |

Learner name: \_\_\_\_\_ Date: \_\_\_\_\_ Mentor feedback:  
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